

Two syringes filled with water are connected by a tube. The first syringe has a plunger with an area of 3 cm^2 , and the second syringe has a plunger with an area of 6 cm^2 . If a force of 10 N is applied to the first syringe, what force acts on the plunger of the second syringe?

- ☐ A. 3 N [1%]
☐ B. 5 N [32%]
☐ C. 10 N [11%]
✓ ☒ D. 20 N [53%]

Correct

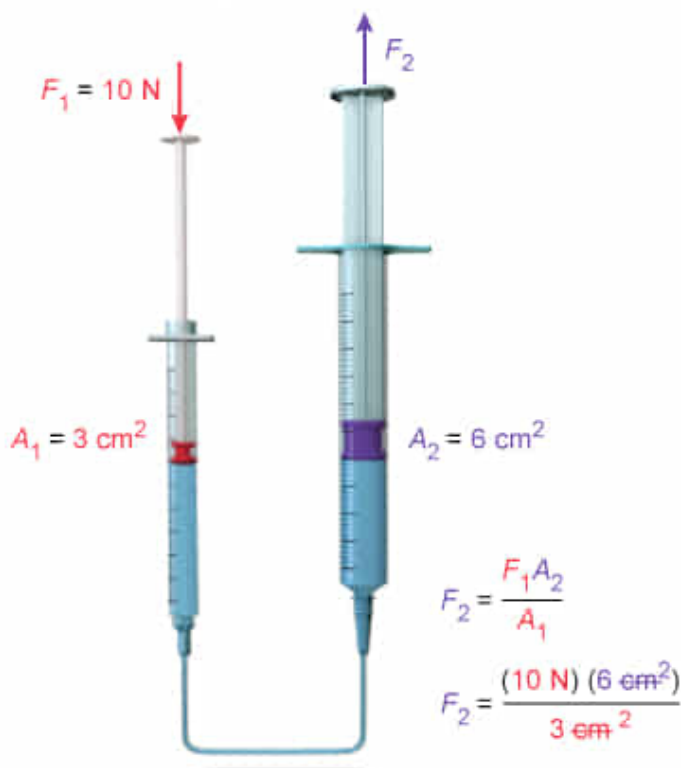
54% answered correctly

Explanation:

Forces on two connected syringes

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

- F_1 Force on first syringe
- A_1 Plunger area of first syringe
- F_2 Force on second syringe
- A_2 Plunger area of second syringe



$$F_2 = 20 \text{ N}$$

Pascal's law states that pressure applied to an **ideal fluid** is uniformly transmitted to all regions of the fluid. **Pressure** P equals the ratio of the **force** F applied and the **area** A over which it acts:

$$P = \frac{F}{A}$$

In this question, the first syringe has a plunger area A_1 of 3 cm^2 and the second syringe has a plunger area A_2 of 6 cm^2 . Because the syringes are connected, they experience the same pressure:

$$P = P_1 = P_2$$

Based on these two equations, the force F_1 on the first syringe and the force F_2 on the second syringe are related by:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Solving this equation for F_2 gives:

$$F_2 = \frac{F_1 A_2}{A_1}$$

Substituting the values for F_1 , A_1 , and A_2 yields:

$$F_2 = \frac{(10 \text{ N})(6 \text{ cm}^2)}{3 \text{ cm}^2} = 20 \text{ N}$$

Therefore, the plunger of the second syringe experiences F_2 of 20 N.

(Choice A) 3 N results from using 1 cm^2 as the plunger area of the second syringe instead of 6 cm^2 .

(Choice B) 5 N is calculated from the ratio of the plunger areas of the first syringe and the second syringe instead of the ratio of the plunger areas of the second syringe and the first syringe.

(Choice C) 10 N would be the force on the second syringe if both syringes had the same plunger area, but the second syringe's plunger area is twice that of the first syringe's plunger area.

Educational objective:

Pascal's law states that an external pressure applied to an ideal fluid is uniformly transmitted to all regions of the fluid. As a result, the ratios of any force and the area to which it is applied must be equal.

Time spent:0 sec

Subject:
Physics

QID:404152

System:
Fluids